

Motion Parameters

Data

The notebook includes data from:

1. Motion parameters (DVARs) derived from running `fsl_motion_outliers` for all subjects
2. Additional motion parameters derived from extracting DVARs and sDVARs (from <https://github.com/nicholst/StandardizedDVARs/blob/master/DVARs.sh>)

Libraries

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/Motion")
motion <- read.csv("MotionData.csv")
```

Check for Outliers

```
# Calculate outlier percentage
Percent_outlier <- (motion$outliers / motion$nvols)*100
motion <- cbind(motion, Percent_outlier)

Outlier <- ifelse(motion$Percent_outlier >20,1,0)
motion <- cbind(motion, Outlier)
rm(Outlier)

# Subjects to exclude from analysis
ID_outliers <- subset(motion, Outlier==1)
unique(ID_outliers$S_ID)

## [1] TIDES_P2T_S4
## 20 Levels: TIDES_P2T_S1 TIDES_P2T_S11 TIDES_P2T_S12 ... TIDES_P2T_S9
```

Import data with motion metrics per stimulation condition - This was done using the script ExtractMotionPerCondition.m

```
MotionCond <- read_csv("MotionPerCond.csv")

# Assign factors
# TI 1:1, TI 1:3 and Sham are during encoding; resp = response period across conditions

MotionCond$metric <- factor(MotionCond$metric)
MotionCond$StimType <- factor(MotionCond$StimType)

# Based on previous identification of outliers excluding TIDES_P2T_S4
MotionCond <- subset(MotionCond, S_ID!="TIDES_P2T_S4")
```

Calculate mean values for reporting

```
MotionCond %>%
  group_by(StimType,metric) %>%
  summarise(average = mean(meanMotion), std = sd(meanMotion))
```

```
## # A tibble: 9 x 4
## # Groups:   StimType [5]
##   StimType metric average    std
##   <fct>    <fct>    <dbl> <dbl>
## 1 mean_all DVARs     25.4  2.63
## 2 mean_all sDVARs     1.06 0.0597
## 3 resp     DVARs     24.7  2.33
## 4 Sham     DVARs     24.7  2.60
## 5 Sham     sDVARs     1.05 0.0581
## 6 TI 1:1   DVARs     25.4  4.01
## 7 TI 1:1   sDVARs     1.07 0.0861
## 8 TI 1:3   DVARs     25.5  3.65
## 9 TI 1:3   sDVARs     1.07 0.0831
```

Calculate DVARs per Stim Condition

```
# Compare DVARs across stim conditions
# Excluding resp period for now - could be compared later to each condition or the average of stim cond
FN.Motion.DVARs <- subset(MotionCond, metric=="DVARs" & StimType!="resp" & StimType!="mean_all")

DVARsfit <- lmer(meanMotion ~ StimType + (1|S_ID), data = FN.Motion.DVARs)
Anova(DVARsfit, test="F")

## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: meanMotion
##           F Df Df.res Pr(>F)
## StimType 1.0297 2    38 0.3669
```

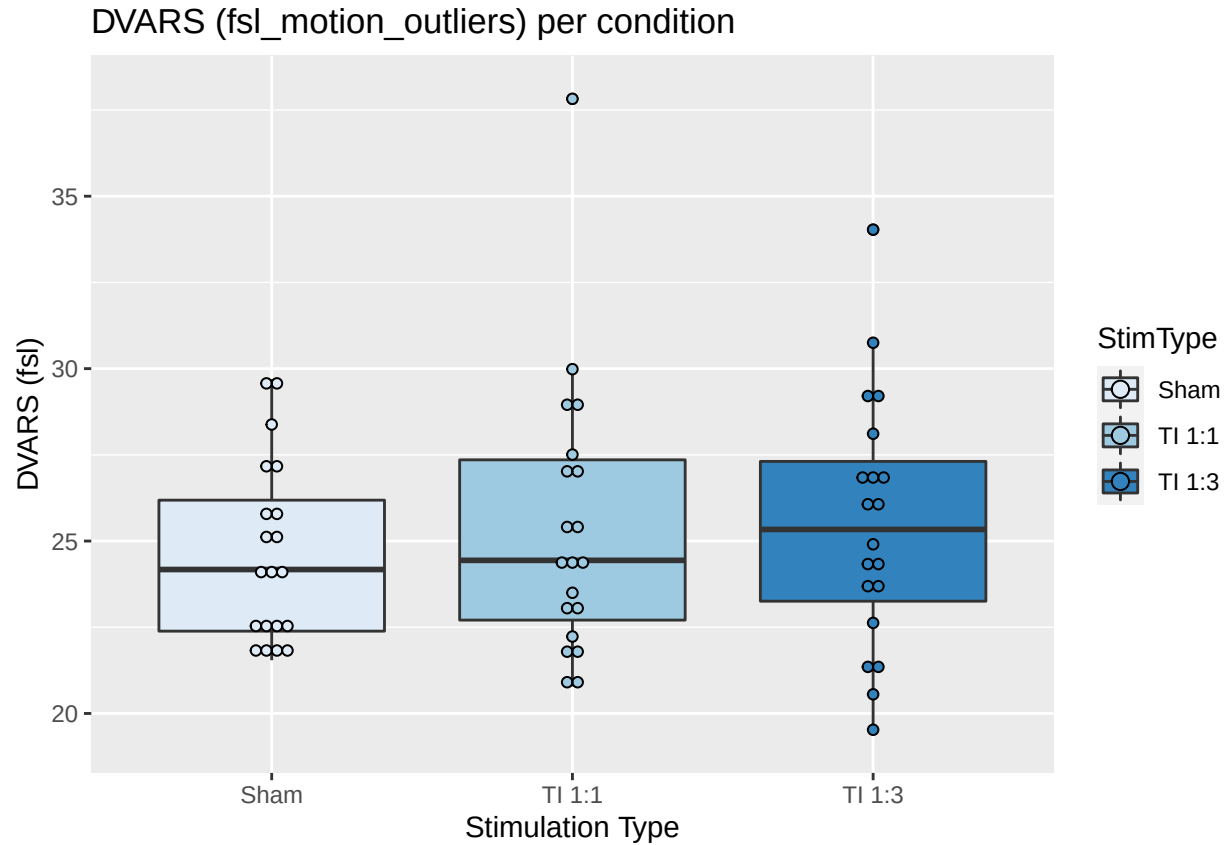
```
summary(DVARsfit, ddf = "Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: meanMotion ~ StimType + (1 | S_ID)
## Data: FN.Motion.DVARs
##
## REML criterion at convergence: 287.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.72660 -0.59652 -0.03908  0.41055  2.94883
##
## Random effects:
## Groups Name Variance Std.Dev.
## S_ID (Intercept) 7.975 2.824
## Residual 4.065 2.016
## Number of obs: 60, groups: S_ID, 20
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 24.6803 0.7759 30.3582 31.809 <2e-16 ***
## StimTypeTI 1:1 0.7446 0.6375 38.0000 1.168 0.250
## StimTypeTI 1:3 0.8327 0.6375 38.0000 1.306 0.199
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) STTI1:1
## StmTypTI1:1 -0.411
## StmTypTI1:3 -0.411 0.500
```

```
estimated_means <- get_means(DVARsfit, "StimType")
pander(estimated_means)
```

StimType	Mean	SE	df	CI_lower	CI_higher
Sham	24.68	0.7759	30.36	23.1	26.26
TI 1:1	25.42	0.7759	30.36	23.84	27.01
TI 1:3	25.51	0.7759	30.36	23.93	27.1

```
boxplot_DVARs <- ggplot(FN.Motion.DVARs, aes(x=StimType, y=meanMotion, fill=StimType)) +
  geom_boxplot() +
  geom_dotplot(binaxis='y', stackdir='center', dotsize = 0.5)
boxplot_DVARs + scale_fill_brewer(palette="Blues") + ggtitle("DVARs (fsl_motion_outliers) per condition")
xlab("Stimulation Type") + ylab("DVARs (fsl)")
```



Calculate sDVARs per Stim Condition

```
# Compare sDVARs across stim conditions
# Excluding resp period for now - could be compared later to each condition or the average of stim cond
FN.Motion.sDVARs <- subset(MotionCond, metric=="sDVARs" & StimType!="resp" & StimType!="mean_all")

sDVARsfit <- lmer(meanMotion ~ StimType + (1|S_ID), data = FN.Motion.sDVARs)
anova.m1 = anova(sDVARsfit)
print(anova.m1)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF DenDF F value Pr(>F)
## StimType  8.3708  4.1854     2    38  1.0297 0.3669
```

```
pander(anova.m1)
```

Table 2: Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
StimType	8.371	4.185	2	38	1.03	0.3669

```
estimated_means <- get_means(sDVARsfit, "StimType")
pander(estimated_means)
```

StimType	Mean	SE	df	CI_lower	CI_higher
Sham	1.046	0.01717	44.3	1.012	1.081
TI 1:1	1.072	0.01717	44.3	1.037	1.106
TI 1:3	1.074	0.01717	44.3	1.039	1.109

```
boxplot_sDVARs <- ggplot(FN.Motion.sDVARs, aes(x=StimType, y=meanMotion, fill=StimType)) +
  geom_boxplot() +
  geom_dotplot(binaxis='y', stackdir='center', dotsize = 0.5)
boxplot_sDVARs + scale_fill_brewer(palette="Blues") + ggtitle("standardise DVARs per condition") +
  xlab("Stimulation Type") + ylab("sDVARs")
```

